weight of the whole tower, a figure close to a staggering 7,00,000 tons in this case.

Mobile connectivity

It's not rare to see lots of people standing next to windows or in the open lobby in multi-storied building hooked to their mobile phones because of poor signal strength in their offices. Imagine this problem at the scale of *Burj Khalifa*. The engineers at task have handled this problem in style. Two local companies Du and Etisalat have provided complete connectivity inside the building including elevators and lobbies at speeds exceeding anywhere else in the region. The building is equipped with state-of-the-art HSPA+ 3.5G mobile connectivity that offers speeds up to 21 Mbps for download and 11 Mbps for uploads.

Elevators

One of the primary things that make a skyscraper practical is the inclusion of elevators. Choosing the number, size and configuration of elevators is a major decision because increasing the number of floors in a building leads to more occupants who need more elevators to carry them, but elevators occupy a lot of valuable floor space and hence decreasing occupancy potential.

The Burj Khalifa, not surprisingly, has the fastest elevator in the world and can travel at a peak speed of 64 kmph – faster than some of the first aircraft. At this speed, you can descend more than 4 stories in a single second. Even the relatively slow elevators in Taipei 101 can transfer you from fifth to 89th floor in just 37 seconds.

Most skyscrapers in order to save space and increase capacity use double-deck elevators where via the same elevator shaft two floors are serviced at the same time. Obvious disadvantages include increased travel time as passengers on both floors have to stop even if the elevator needs to stop at only one floor. *Burj Khalifa* is also one of the few buildings in the world in which certain elevators can be used for speedy evacuation in case of fire as opposed to conventional scenarios where elevator operation is stopped during fire. The building service/fireman's elevator will have a capacity of 5,500 kg, compare this to the elevator in your building which generally has a capacity of 800 kg or less.

Maintenance

Simple and trivial as it may look, maintenance work post-construction in itself is a major challenge. Take the example of cleaning the windows in *Burj Khalifa*. To accomplish this seemingly simple task, a system worth more than $8 million (approximately Rs. 40 crore) is installed in Burj Khalifa by an Australian company called Cox Gomyl. The machinery was manufactured in Airport West and is operated in manned mode for the lower stories and completely unmanned after that. It takes about 3 months to complete the whole building, so yes, as you can imagine, it is dirty by the end of a cycle and the system is operated continuously.

So you thought cleaning your windows was tough?

Horizontal tracks are fixed on the exterior via which small transport hubs move carrying workers and equipment, when not in use they are tucked away out of sight to maintain aesthetics. The tracks are located on the level number 40, 73 and 1,009. Each track is equipped with a 1.5 tonne bucket machine which can slide over the tracks horizontally and then will descend vertically using strong cables to cover the whole building.

Aesthetics

A seemingly unimportant consideration on the surface, but having the tallest building in your country has become an issue of national pride. It is similar to hosting the Olympics or Commonwealth games and exhibits the technological and economic advancements of your nation. The task of ensuring that such a symbol of national pride is aesthetically appealing is taken very seriously by the designers and can cost a major fraction of the total time and money spent on the project.

For example, there are over 1,000 pieces of art that will be on display inside the *Burj Khalifa*. The lobby alone boasts of 196 cymbals constructed out of bronze and brass alloy, each representing one nation of the world. They are plated by 18-carat gold and individually produce a distinctive sound when struck by dripping water, intended to mimic the sound of water falling on leaves.

On the outside stands an electronically controlled fountain designed by WET Design (the California-based company whose other installations include fountains at the Bellagio Hotel Lake in Las Vegas) at a cost of $217 million (over Rs. 1,000 crore). The fountain is fitted with over 6,000 lights, each of which can be controlled electronically and has 50 coloured projectors. The 275-metre long fountain can shoot water up to a height of 150 metres (that is over 40 stories, so yes if you are reading this article anywhere in India, the fountain can drench you on your rooftop) is a technological marvel in itself.

The race to build the tallest and strongest includes the use of wind tunnels, material research which ironically involves use of nano particles to form stable mega structures, sophisticated digital monitoring systems, automated cranes and so on. As cities get ever more crowded the next natural step is to construct building over 1 kilometre in height and you might not have to hold your breath far too long for it since the Saudi Prince Al Waleed bin Talal is already on the job.

If you don't know why this is here, turn to page 109 to start playing Digit's Crack The Code contest
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